

Your Paper Title Here

Author Name¹ and Co-Author Name^{2†}

¹Department of Mechanical Science and Engineering, University of Illinois
Urbana–Champaign, 1206 West Green Street, Urbana, IL 61801, USA

²Department of Aerospace Engineering, University of Illinois Urbana–Champaign, 104 South
Wright Street, Urbana, IL 61802, USA

(Received xx; revised xx; accepted xx)

Your abstract goes here.

Key words: Authors should not enter keywords on the manuscript, as these must be chosen by the author during the online submission process and will then be added during the typesetting process (see <http://journals.cambridge.org/data/relatedlink/jfm-keywords.pdf> for the full list)

1. Introduction

Your introduction goes here.

2. Governing equations and numerical methods

2.1. *Governing equations*

Your governing equations go here.

2.2. *Objective function and adjoint method*

Your objective function and adjoint method description goes here.

2.3. *Numerical methods*

Your numerical methods description goes here.

† Email address for correspondence: email@address.edu

3. Results and discussion

Your results go here.

3.1. Results

Your results go here.

3.2. Discussion

Your discussion goes here.

4. Conclusions

Your conclusions go here.

Appendix A. Appendix

A.1. Derivation details

Your derivation details go here.

A.2. Additional results

Your additional results go here.

REFERENCES

- ALVARADO, JOSÉ, TEICH, ERIN, SIVAK, DAVID & BECHHOEFER, JOHN 2025 Optimal control in soft and active matter. *arXiv preprint arXiv:2504.08676* .
- BEWLEY, THOMAS R, MOIN, PARVIZ & TEMAM, ROGER 2001 Dns-based predictive control of turbulence: an optimal benchmark for feedback algorithms. *Journal of Fluid Mechanics* **447**, 179–225.
- BINGHAM, CHRISTOPHER 1974 An antipodally symmetric distribution on the sphere. *The Annals of Statistics* pp. 1201–1225.
- BUTTINONI, IVO, BIALKÉ, JULIAN, KÜMMEL, FELIX, LÖWEN, HARTMUT, BECHINGER, CLEMENS & SPECK, THOMAS 2013 Dynamical clustering and phase separation in suspensions of self-propelled colloidal particles. *Physical review letters* **110** (23), 238301.
- CHAUBAL, CHARU V & LEAL, L GARY 1998 A closure approximation for liquid-crystalline polymer models based on parametric density estimation. *Journal of Rheology* **42** (1), 177–201.
- CHUNG, SEUNG WHAN & FREUND, JONATHAN B 2022 An optimization method for chaotic turbulent flow. *Journal of Computational Physics* **457**, 111077.

- CICHOS, FRANK, GUSTAVSSON, KRISTIAN, MEHLIG, BERNHARD & VOLPE, GIOVANNI 2020 Machine learning for active matter. *Nature Machine Intelligence* **2** (2), 94–103.
- COLEN, JONATHAN, HAN, MING, ZHANG, RUI, REDFORD, STEVEN A, LEMMA, LINNEA M, MORGAN, LINK, RUIJGROK, PAUL V, ADKINS, RAYMOND, BRYANT, ZEV, DOGIC, ZVONIMIR & OTHERS 2021 Machine learning active-nematic hydrodynamics. *Proceedings of the National Academy of Sciences* **118** (10), e2016708118.
- DOMBROWSKI, CHRISTOPHER, CISNEROS, LUIS, CHATKAEW, SUNITA, GOLDSTEIN, RAYMOND E & KESSLER, JOHN O 2004 Self-concentration and large-scale coherence in bacterial dynamics. *Physical review letters* **93** (9), 098103.
- EZHILAN, BARATH, SHELLEY, MICHAEL J & SAINTILLAN, DAVID 2013 Instabilities and nonlinear dynamics of concentrated active suspensions. *Physics of Fluids* **25** (7), 070607.
- FALK, MARTIN J, ALIZADEHYAZDI, VAHID, JAEGER, HEINRICH & MURUGAN, ARVIND 2021 Learning to control active matter. *Physical Review Research* **3** (3), 033291.
- FRANGIPANE, GIACOMO, DELL’ARCIPRETE, DARIO, PETRACCHINI, SERENA, MAGGI, CLAUDIO, SAGLIMBENI, FILIPPO, BIANCHI, SILVIO, VIZSNYICZAI, GASZTON, BERNARDINI, MARIA LINA & DI LEONARDO, ROBERTO 2018 Dynamic density shaping of photokinetic e. coli. *Elife* **7**, e36608.
- FREUND, JONATHAN B 2023 Object transport by a confined active suspension. *Journal of Fluid Mechanics* **960**, A16.
- GAO, TONG, BETTERTON, MEREDITH D, JHANG, AN-SHENG & SHELLEY, MICHAEL J 2017 Analytical structure, dynamics, and coarse graining of a kinetic model of an active fluid. *Physical Review Fluids* **2** (9), 093302.
- GOMPPER, GERHARD, WINKLER, ROLAND G, SPECK, THOMAS, SOLON, ALEXANDRE, NARDINI, CESARE, PERUANI, FERNANDO, LÖWEN, HARTMUT, GOLESTANIAN, RAMIN, KAUPP, U BENJAMIN, ALVAREZ, LUIS & OTHERS 2020 The 2020 motile active matter roadmap. *Journal of Physics: Condensed Matter* **32** (19), 193001.
- KAISER, ANDREAS, PESHKOV, ANTON, SOKOLOV, ANDREY, TEN HAGEN, BORGE, LÖWEN, HARTMUT & ARANSON, IGOR S 2014 Transport powered by bacterial turbulence. *Physical review letters* **112** (15), 158101.
- KLEIN, BRANDON, FRANCO, ALEJANDRO J SOTO, SABBIR, MD MAINUL HASAN, DEUTSCH, MATTHEW J, KLEGMAN, ROSS, SELINGER, ROBIN LB, MITCHELL, KEVIN A & BELLER, DANIEL A 2025 Spontaneous optimal mixing via defect-vortex coupling in confined active nematics. *arXiv preprint arXiv:2503.10880* .
- KOUMAKIS, N, LEPORE, A, MAGGI, C & DI LEONARDO, R 2013 Targeted delivery of colloids by swimming bacteria. *Nature communications* **4** (1), 2588.
- LAMTYUGINA, ALEXANDRA, QIU, YUQING, FODOR, ÉTIENNE, DINNER, AARON R & VAIKUNTANATHAN, SURIYANARAYANAN 2022 Thermodynamic control of activity patterns in cytoskeletal networks. *Physical review letters* **129** (12), 128002.
- LEMMMA, LINNEA M, VARGHESE, MINU, ROSS, TYLER D, THOMSON, MATT, BASKARAN,

- APARNA & DOGIC, ZVONIMIR 2023 Spatio-temporal patterning of extensile active stresses in microtubule-based active fluids. *PNAS nexus* **2** (5), pgad130.
- MARCHETTI, M CRISTINA, JOANNY, JEAN-FRANÇOIS, RAMASWAMY, SRIRAM, LIVERPOOL, TANNIEMOLA B, PROST, JACQUES, RAO, MADAN & SIMHA, R ADITI 2013 Hydrodynamics of soft active matter. *Reviews of modern physics* **85** (3), 1143.
- MASSANA-CID, HELENA, MAGGI, CLAUDIO, FRANGIPANE, GIACOMO & DI LEONARDO, ROBERTO 2022 Rectification and confinement of photokinetic bacteria in an optical feedback loop. *Nature Communications* **13** (1), 2740.
- NEEDLEMAN, DANIEL & DOGIC, ZVONIMIR 2017 Active matter at the interface between materials science and cell biology. *Nature reviews materials* **2** (9), 1–14.
- NORTON, MICHAEL M, GROVER, PIYUSH, HAGAN, MICHAEL F & FRADEN, SETH 2020 Optimal control of active nematics. *Physical review letters* **125** (17), 178005.
- PARTOVIFARD, ARGHAVAN, GRAWITTER, JOSUA & STARK, HOLGER 2024 Controlling active turbulence by activity patterns. *Soft Matter* **20** (8), 1800–1814.
- QUAH, TITUS, TAKATORI, SHO C & RAWLINGS, JAMES B 2025 Physics-informed neural model predictive control of interacting active brownian particles. *arXiv preprint arXiv:2501.18809* .
- SAINTILLAN, DAVID & SHELLEY, MICHAEL J 2008 Instabilities and pattern formation in active particle suspensions: kinetic theory and continuum simulations. *Physical Review Letters* **100** (17), 178103.
- SANCHEZ, TIM, CHEN, DANIEL TN, DECAMP, STEPHEN J, HEYMANN, MICHAEL & DOGIC, ZVONIMIR 2012 Spontaneous motion in hierarchically assembled active matter. *Nature* **491** (7424), 431–434.
- SAW, THUAN BENG, DOOSTMOHAMMADI, AMIN, NIER, VINCENT, KOCGOZLU, LEYLA, THAMPI, SUMESH, TOYAMA, YUSUKE, MARCQ, PHILIPPE, LIM, CHWEE TECK, YEOMANS, JULIA M & LADOUX, BENOIT 2017 Topological defects in epithelia govern cell death and extrusion. *Nature* **544** (7649), 212–216.
- SHELLEY, MICHAEL J 2016 The dynamics of microtubule/motor-protein assemblies in biology and physics. *Annual review of fluid mechanics* **48**, 487–506.
- SOKOLOV, ANDREY, APODACA, MARIO M, GRZYBOWSKI, BARTOSZ A & ARANSON, IGOR S 2010 Swimming bacteria power microscopic gears. *Proceedings of the National Academy of Sciences* **107** (3), 969–974.
- SORIANI, ARTUR, TJHUNG, ELSSEN, FODOR, ÉTIENNE & MARKOVICH, TOMER 2025 Control of active field theories at minimal dissipation. *arXiv preprint arXiv:2504.19285* .
- TAKATORI, SHO C, QUAH, TITUS & RAWLINGS, JAMES B 2025 Feedback control of active matter. *Annual Review of Condensed Matter Physics* **16** (1), 319–341.
- TURIV, TARAS, KRIEGER, JESS, BABAKHANOVA, GRETA, YU, HAO, SHIYANOVSKII, SERGIJ V, WEI, QI-HUO, KIM, MIN-HO & LAVRENTOVICH, OLEG D 2020 Topology control of human fibroblast cells monolayer by liquid crystal elastomer. *Science Advances* **6** (20), eaaz6485.

- VISHNAMPET, RAMANATHAN, BODONY, DANIEL J & FREUND, JONATHAN B 2015 A practical discrete-adjoint method for high-fidelity compressible turbulence simulations. *Journal of Computational Physics* **285**, 173–192.
- VIZSNYICZAI, GASZTON, FRANGIPANE, GIACOMO, MAGGI, CLAUDIO, SAGLIMBENI, FILIPPO, BIANCHI, SILVIO & DI LEONARDO, ROBERTO 2017 Light controlled 3d micromotors powered by bacteria. *Nature communications* **8** (1), 15974.
- WALTER, JESSICA M, GREENFIELD, DEREK, BUSTAMANTE, CARLOS & LIPHARDT, JAN 2007 Light-powering escherichia coli with proteorhodopsin. *Proceedings of the National Academy of Sciences* **104** (7), 2408–2412.
- WEADY, SCOTT, STEIN, DAVID B & SHELLEY, MICHAEL J 2022 Thermodynamically consistent coarse-graining of polar active fluids. *Physical Review Fluids* **7** (6), 063301.
- WEI, MINGJUN & FREUND, JONATHAN B 2006 A noise-controlled free shear flow. *Journal of Fluid Mechanics* **546**, 123–152.
- ZHANG, RUI, REDFORD, STEVEN A, RUIJGROK, PAUL V, KUMAR, NITIN, MOZAFFARI, ALI, ZEMSKY, SASHA, DINNER, AARON R, VITELLI, VINCENZO, BRYANT, ZEV, GARDEL, MARGARET L & OTHERS 2021 Spatiotemporal control of liquid crystal structure and dynamics through activity patterning. *Nature materials* **20** (6), 875–882.
- ZHOU, ZHENGYANG, JOSHI, CHAITANYA, LIU, RUOSHI, NORTON, MICHAEL M, LEMMA, LINNEA, DOGIC, ZVONIMIR, HAGAN, MICHAEL F, FRADEN, SETH & HONG, PENGYU 2021 Machine learning forecasting of active nematics. *Soft matter* **17** (3), 738–747.